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# StayNest: Guest Property Booking and Reservation Management

- HOW THE PROCESS BEGINS AND PROGRESSES:

The process starts when a guest opens StayNest and registers or logs in. The guest searches by location, price and dates, and the system returns matching properties from the database. When the guest picks a property, the system checks its availability calendar. If the dates are taken, the guest chooses other dates or another property. If they are free, the guest enters the booking details, which the system validates and checks for availability a second time to avoid conflicts. A valid request is saved with the status PENDING and the host is notified.

The host reviews the request and approves or rejects it. A rejection updates the status and notifies the guest. After an approval the guest pays. The payment system processes online payments, or an administrator verifies a bank or mobile-money reference. Once the payment is VERIFIED, the booking becomes CONFIRMED, the dates are locked for other guests, and the guest and host receive confirmations. The process then ends, unless a cancellation is requested later.

- WHAT EACH ACTOR DOES:

**Guest:** registers, searches, books, pays and follows the booking status. **Host:** owns the properties, reviews requests and approves or rejects them. **Administrator:** supervises bookings, verifies manual payments, handles authorized cancellations and prepares reports. **Payment System:** processes transactions and returns a reference and a status. **Notification System:** sends every message: request, approval, rejection, payment update, confirmation and cancellation.

- HOW DECISSIONS ARE MADE:

Five decisions drive the flow. (1) Is the property available? (2) Is the booking information valid? (3) Does the host approve? (4) Is the payment confirmed? (5) Is a cancellation requested and permitted? Each answer is a rule on stored data: availability comes from CONFIRMED bookings and the property's calendar flag, validity from the booking rules, and the other three from a recorded status. A "no" sends the process back to the guest or leaves the booking in a pending status.

- HOW INFORMATIONS MOVES:

The database is the single shared record. The guest sends search criteria, dates and payment details. The system reads properties and bookings, and writes bookings, payments and notifications. The host receives request details and returns a decision. The payment system returns a reference and a status. The administrator reads bookings, payments, and writes verification and cancellation updates. In the diagram, each activity is marked as Create, Read, Update or Delete/Archive: for example, registration, booking requests, payments and notifications create records; search and availability checks read them; approval, verification, confirmation and cancellation update them; and completed or cancelled bookings are archived, not erased.

- HOW THE PROCESS SUPPORTS OPERATIONS AND MANAGEMENT:

Operationally, the process reduces double bookings, speeds up communication between guests and hosts, and keeps every request, payment and status traceable. For management, the stored data answers questions such as how many bookings are pending, confirmed or cancelled, which properties and neighborhoods earn the most, how much revenue is verified per host, and how often bookings are cancelled. Administrators use these figures to monitor performance, spot payment problems and decide on pricing, host support and growth.